

Derivation of the Volume Formula for Square and Rectangular Pyramids

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A **pyramid** is a solid figure with one **polygonal base** and triangular faces meeting at a common **vertex**.

Square and Rectangular Pyramids

A **square pyramid** has a square base, while a **rectangular pyramid** has a rectangular base.

Volume of a Pyramid

The **volume of a pyramid** is one-third of the volume of a prism with the same base and height.

$$V = \frac{1}{3}Bh$$

where B is the **area of the base** and h is the **height**.

Square Pyramid

For a square pyramid,

$$V = \frac{1}{3}s^2h$$

where s is the side length of the square base.

Rectangular Pyramid

For a rectangular pyramid,

$$V = \frac{1}{3}lwh$$

where l is the length, w is the width, and h is the height.

Examples

Example 1

A square pyramid has a base with side length 6 cm and a height of 9 cm. Find its volume.

Example 2

A square pyramid has a base area of 81 cm^2 and a height of 15 cm . Find its volume.

Example 3

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 9 cm. Find its volume.

Practice Exercises

Practice 1

Find the volume of the pyramid.

A square pyramid has a base with side length 6 cm and a height of 9 cm.

Practice 2

Find the volume of the pyramid.

A square pyramid has a base area of 81 cm^2 and a height of 15 cm .

Practice 3

Find the volume of the pyramid.

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 9 cm.

Practice 4

Find the volume of the pyramid.

A square pyramid has side length 12 m and height 10 m.

Practice 5

Find the volume of the pyramid.

A rectangular pyramid has dimensions 14 cm by 9 cm with height 12 cm.

Practice 6

Find the volume of the pyramid.

A pyramid has a rectangular base with area 96 cm^2 and height 15 cm .

Activities

Activity 1

Find the volume of the pyramid.

A square pyramid has a base with side length 8 cm and a height of 12 cm.

Activity 2

Find the volume of the pyramid.

A square pyramid has a base area of 100 cm^2 and a height of 18 cm.

Activity 3

A prism has a volume of 720 cm^3 . A pyramid has the same base and height as the prism. Find the volume of the pyramid.

Activity 4

Find the volume of the pyramid.

A square pyramid has side length 15 m and height 8 m.

Activity 5

Find the volume of the pyramid.

A rectangular pyramid has dimensions 18 cm by 10 cm with height 9 cm.

Activity 6

Find the volume of the pyramid.

A pyramid has a rectangular base with area 150 cm^2 and height 14 cm .

Thank you!